Deep Learning class assignments

Each project is expected to take up to a month and will be individually reviewed during project meetings.

Use PyTorch and train in Google Colab (or personal PC if it is capable enough, but a strong GPU is suggested).

1. Implement Highway Network and train it on the CIFAR10 dataset. Read and understand the corresponding article:

https://arxiv.org/pdf/1505.00387

2. Implement ResNET 56 and train it on the CIFAR10 dataset. Read and understand the corresponding article:

https://arxiv.org/abs/1512.03385

3. Implement a complete GPT-2 system (i.e. training and autoregressive inference). Implement with two types of attention layer: your own multi-head attention implementation and PyTorch build-in attention. Compare their speed of training and performance.

The dataset to use is: Tiny Shakespeare.

Read and understand the corresponding article:

https://arxiv.org/abs/1706.03762

4. Implement and train Stable Diffusion.

The SD implementation must use pretrained components:

Pretrained VAE: stabilityai/sd-vae-ft-mse
Pretrained CLIP text encoder: openai/clip-vit-large-patch14

The dataset: AlekseyKorshuk/celeba-hq-blip

Read and understand the corresponding article:

https://arxiv.org/pdf/2112.10752